Summary in Graph

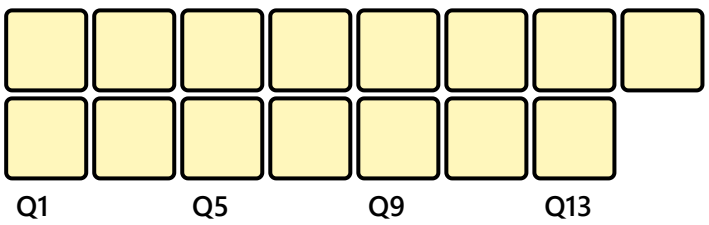
Exam Summary (GO Classes GATE CS | First Order Logic| Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (6 + 9)
Total Marks:	24 (6 + 18)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (62) out of 622 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Mathematical Logic

A binary operator is defined as follows

$$P \Updownarrow Q = \sim P \wedge Q$$

Which of the following statement is equivalent to $P \rightarrow Q$

A. $\sim P \Updownarrow Q$
B. $\sim (P \Updownarrow Q)$
C. $\sim (\sim P \Updownarrow Q)$
D. $\sim (\sim P \Updownarrow \sim Q)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Mathematical Logic

One approach to handling fuzzy logic data might be to design a computer using ternary $(\text{base-}3)$ logic so that data could be stored as "true," "false," and "unknown." If each ternary logic element is called a flit, minimum how many flits are required to represent at least 256 different values?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Consider the following predicates.

- $\text{Rabbit}(x) = x$ is a rabbit.
- $\text{Cute}(x) = x$ is cute.

Consider the following statement E, where the domain of every variable is set of all animals in a jungle J.

$$\text{E} = \forall x(\text{Rabbit}(x) \wedge \text{Cute}(x))$$

If statement E is true, then which of the following is true?

A. There is no animal other than rabbits in the jungle J.
B. Every rabbit is cute in jungle J.
C. It is possible that there is some animal in J who is not a rabbit but is cute.
D. There is some rabbit who is cute in jungle J.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Mathematical Logic

Let $p(x), q(x)$ denote the following open statements.

$$p(x): \quad x \leq 3 \quad q(x): \quad x+1 \text{ is odd}$$

Let $r(x)$ be the open statement " $x > 0$." The universe comprises all integers. Determine the number of values of x for which $[p(x) \wedge q(x)] \wedge r(x)$ results in a true statement.

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Consider the following predicates:

- $\text{bird}(x)$: x is a bird.
- $\text{penguin}(x)$: x is a penguin.

- $\text{fly}(x)$: x can fly.

Consider the following sentence:

All birds except penguins' fly. (Note that Penguins may or may not fly)

Which of the following is/are a formula in first-order logic expressing the given fact?

A. $\forall x.\ [(\text{bird}(x) \wedge \neg \text{ penguin } (x)) \rightarrow \text{fly}(x)]$
B. $\forall x.\ [\neg \text{ bird}(x) \vee \neg \text{ penguin } (x) \vee \text{ fly } (x)]$
C. $\forall x.\ [\neg \text{ bird } (x) \vee \text{ penguin } (x) \vee \text{ fly } (x)]$
D. $\forall x.\ [\text{bird}(x) \vee \text{ penguin } (x) \vee \text{ fly } (x)]$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Consider the following English sentence:

"Agra and Gwalior are both in India".

A student has written a logical sentence for the above English sentence in First-Order Logic using predicate $\text{IN}(x, y)$, which means x is in y, as follows.

$\text{In}(\text{Agra, India}) \vee \text{In}(\text{Gwalior, India})$

Which one of the following is correct with respect to the above logical sentence?

A. It is syntactically valid but does not express the meaning of the English sentence
B. It is syntactically valid and expresses the meaning of the English sentence also
C. It is syntactically invalid but expresses the meaning of the English sentence
D. It is syntactically invalid and does not express the meaning of the English sentence

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 02sec | Discuss

Q #7

Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $p(x), q(x)$, and $r(x)$ denote the following open statements.

$$\begin{aligned} p(x): &\quad x^2 - 8x + 15 = 0 \\ q(x): &\quad x \text{ is odd} \\ r(x): &\quad x > 0 \end{aligned}$$

For the universe of all integers, which of the following statements is/are false?

A. $\forall x[p(x) \rightarrow q(x)]$
B. $\forall x[q(x) \rightarrow p(x)]$
C. $\exists x[p(x) \rightarrow q(x)]$
D. $\exists x[q(x) \rightarrow p(x)]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8

Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $p(x), q(x)$, and $r(x)$ denote the following open statements.

$$p(x): \quad x^2 - 8x + 15 = 0$$
$$q(x): \quad x \text{ is odd}$$
$$r(x): \quad x > 0$$

For the universe of all integers, which of the following statements is/are true?

A. $\exists x[r(x) \rightarrow p(x)]$
B. $\forall x[\neg q(x) \rightarrow \neg p(x)]$
C. $\exists x[p(x) \rightarrow (q(x) \wedge r(x))]$
D. $\forall x[(p(x) \vee q(x)) \rightarrow r(x)]$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Which of the following formulas is a formalization of the sentence: "No dogs are intelligent"

A. $\forall \mathrm{x}(\mathrm{dog}(\mathrm{x}) \wedge \neg \mathrm{Intelligent}(\mathrm{x}))$
B. $\forall \mathrm{x}(\mathrm{Intelligent}(\mathrm{x}) \rightarrow \neg \mathrm{dog}(\mathrm{x}))$
C. $\forall x(\mathrm{dog}(\mathrm{x}) \rightarrow \neg \mathrm{Intelligent}(\mathrm{x}))$
D. $\neg \exists x(\mathrm{dog}(\mathrm{x}) \wedge \mathrm{Intelligent}(\mathrm{x}))$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Which of the following formulas is a formalization of the sentence : "All babies are illogical"

A. $\forall \mathrm{x}(\mathrm{baby}(\mathrm{x}) \wedge \mathrm{illogical}(\mathrm{x}))$
B. $\forall \mathrm{x}(\mathrm{baby}(\mathrm{x}) \rightarrow \mathrm{illogical}(\mathrm{x}))$
C. $\neg \exists \mathrm{x}(\mathrm{baby}(\mathrm{x}) \wedge \neg \mathrm{illogical}(\mathrm{x}))$
D. $\exists \mathrm{x}(\mathrm{baby}(\mathrm{x}) \wedge \neg \mathrm{illogical}(\mathrm{x}))$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Consider the following predicates:

- sqroot((x) : x has a square root.
- negative (x) : x is negative.

Which of the following formulas is a formalization of the sentence, if the domain is set of all numbers: "Every number is either negative or has a square root but not both"

A. $\forall \mathrm{x} \neg($ negative $(\mathrm{x}) \leftrightarrow \mathrm{sqroot}(\mathrm{x}))$
B. $\neg \exists \mathrm{x}((\text{negative } (\mathrm{x}) \leftrightarrow \text{sqroot } (\mathrm{x}))$
C. $\forall \mathrm{x}((\text{ negative } (\mathrm{x}) \wedge \neg \mathrm{sqroot}(\mathrm{x})) \vee (\neg \text{ negative } (\mathrm{x}) \wedge \mathrm{sqroot}(\mathrm{x})))$
D. $\forall \mathrm{x}(\text{ negative } (\mathrm{x}) \vee \mathrm{sqroot}(\mathrm{x}))$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Consider the following predicates:

- $\text{Real}(x)$: x is a real number.
- $\text{Integer}(x)$: x is an integer.

Which of the following formulas is a formalization of the sentence, if the domain is set of all integers. "Some integers are not real numbers."

A. $\exists x \neg \text{Real}(x)$
B. $\neg \forall x\, \text{Real}(x)$
C. $\exists x(\text{Integer}(x) \wedge \neg \text{Real}(x))$
D. $\exists x(\text{Integer}(x) \rightarrow \neg \text{Real}(x))$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Which of the following formulas is a formalization of the sentence :

Not all that glitters is gold

A. $\forall x$ [glitter(x) $\rightarrow \text{gold}(x)]$
B. $\neg \exists x[\text{glitter}(x) \wedge \text{gold}(x)]$
C. $\neg \forall x(\text{glitter}(x) \rightarrow \text{gold}(x))$
D. $\exists x(\text{glitter}(x) \wedge \neg \text{gold}(x))$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $P(x), Q(x), R(x)$ and $S(x)$ denote the following predicates with domain $\mathbb{Z}$:

$$\begin{aligned} P(x) &: x \leq 0, \\ Q(x) &: x^2 = 1, \\ R(x) &: x \text{ is odd}, \\ S(x) &: x = x + 1. \end{aligned}$$

Which of the following statements is/are true?

A. $\forall x \in \mathbb{Z}, \quad R(x) \rightarrow S(x)$
B. $\forall x \in \mathbb{Z}, \quad S(x) \rightarrow R(x)$
C. $\exists x \in \mathbb{Z}$ such that $Q(x) \wedge \sim R(x)$
D. $\exists x \in \mathbb{Z}$ such that $P(x) \rightarrow S(x)$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

For a given predicate $\mathrm{P}(\mathrm{x})$, you might believe that the statements $\forall x \mathrm{P}(\mathrm{x})$ or $\exists \mathrm{x} \mathrm{P}(\mathrm{x})$ are either true or false. $\exists x(\neg P(x))$ is false if and only if

A. $\exists x(P(x))$ is true.
B. $\forall x(P(x))$ is true.
C. $\forall x(\neg P(x))$ is true.
D. $\forall x(\neg P(x))$ is false.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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